DANIEL CORREA ROSALES

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PROFILE SUMMARY

Erasmus Mundus MSc graduate in Nanoscience and Nanotechnology (**KU Leuven · Universitat de Barcelona**), specializing in Nanomedicine and Nanotheranostics with thesis research at **IQAC-CSIC**. Hands-on expertise in **peptide amphiphile synthesis** (Fmoc-SPPS), **drug encapsulation systems**, **nanoscale instrumentation** (AFM, Raman, HPLC, TEM/HRTEM, ESI-MS), and **quantitative data analysis** (Python, molecular dynamics, statistical modeling). **3 years professional backend software engineering** in reactive Java microservices at one of Latin America's largest banks; core contributor to team-led microservices redesign, developing services from scratch that enabled production scaling to **5,000+ account creations/day**, with subsequent architectural extensions supporting specialized account types at **additional 1,000+ daily**. Physics BSc with strong foundation in semiconductor device physics and materials science. Uniquely positioned to bridge research-grade nanoscience with robust software development for instrumentation-driven, data-intensive, or biomedical technology environments. Available full-time **immediately**.

EDUCATION

Erasmus Mundus MSc in Nanoscience and Nanotechnology

KU Leuven (Year 1) · Universitat de Barcelona (Year 2) *Sept 2024 – Aug 2026*
Leuven, Belgium · Barcelona, Spain

Specialization: Nanomedicine and Nanotheranostics
Thesis: Nanosystems based on peptide amphiphiles for drug encapsulation
Conducted at IQAC-CSIC (Institut de Química Avançada de Catalunya) — CSIC research institute partnered with Universitat de Barcelona
Relevant coursework: Semiconductor Devices · MEMS & Microsystems · Materials Physics for Nanoelectronics · Technology of Integrated Systems

- Synthesized four NCAM-derived peptide amphiphiles with tunable hydrophilic/hydrophobic architectures for ocular drug delivery applications using Fmoc-based solid-phase peptide synthesis (SPPS)
- Confirmed molecular identity through UHPLC coupled with **electrospray ionization mass spectrometry (ESI-MS)** and assessed purity/hydrophobicity trends using reverse-phase HPLC
- Investigated peptide self-assembly through **circular dichroism spectroscopy**, identifying concentration-independent β-sheet secondary structures associated with ordered supramolecular nanostructures
- Determined critical aggregation concentration (CAC) using Thioflavin-T fluorescence assays; demonstrated how arginine incorporation and alkyl-tail number influence self-assembly thermodynamics
- Achieved **encapsulation efficiencies above 95%** for latanoprost-loaded peptide amphiphile nanocarriers and established structure-property relationships governing drug incorporation
- Interpreted the balance between hydrophobic collapse, electrostatic interactions, and β-sheet stabilization in controlling peptide amphiphile aggregation behaviour
- Operated AFM in contact mode to acquire force–distance curves on Si and PDMS; extracted Young's moduli by fitting Hertz/Sneddon contact mechanics models, achieving <1% error vs. literature values (Si: 163 GPa, PDMS: 1–3 MPa)
- Performed AFM nanolithography on gelatin substrates to pattern sub-micron features; controlled hole depth (∼30 nm) by tuning setpoint force (20 nN imaging vs. 1.18 μN patterning) — directly analogous to process parameter optimization in a FAB environment
- Characterized MOF IF142 resistive CO_2 sensor across 11 humidity conditions (2–91% RH); built Python data pipeline to automate IV curve fitting and extract resistance vs. RH with $R^2 = 0.996$ linear model
- Ran 20 ns molecular dynamics simulation of chignolin peptide (AMBER); performed RMSD trajectory clustering (6 clusters, 2,000 frames) and identified dominant folded conformation at 39.6% population fraction
- Conducted quantitative Raman/AFM characterization of fluorographene nanomaterials.
- Completed hands-on **SEM/TEM lab practicals**: prepared samples, observed instrument operation by the lab instructor, analyzed imaging **artifacts**, and practiced SEM focusing and alignment on a **SEM simulator**
- SEM and TEM sample preparation,
- Completed coursework in **Semiconductor Devices**, **MEMS & Microsystems**, **Materials Physics for Nanoelectronics**, and **Technology of Integrated Systems** — providing direct grounding in the physics underlying FAB process areas
- **Attendee — Student Excellence Day, imec (Oct 2025, Leuven)**

B.S. Physics

Yachay Tech University *Sept 2015 – Nov 2021*
Urcuquí-Imbabura, Ecuador

GPA: 3.5 / 4.0

- Thesis: **Basic Molecular Electronics via Mechanically Controlled Break Junction (MCBJ)** Read more
- Operated MCBJ to measure single-molecule conductance; fabricated sub-nm electrode gaps requiring sub-Ångström mechanical precision
- Correlated experimental conductance traces with computed transmission spectra

WORK EXPERIENCE

Production Systems Engineer

Banco Pichincha — one of Latin America's largest banks *May 2022 – Sept 2024*
Quito, Ecuador

- Participated in team-led migration from inefficient microservices architecture to optimized Spring Boot/WebFlux reactive design; developed microservices from scratch. Post-migration system scaled to **5,000+ account creations/day**. Subsequently extended architecture with new account type services, enabling **additional 1,000+ specialized accounts/day**
- Implemented **Java Spring Boot** and **Spring WebFlux** following Clean Code, Clean Architecture, and SOLID principles
- Integrated microservices with Salesforce, PostgreSQL, MongoDB, and Redis
- Achieved **85%+ unit test coverage** using JUnit and Mockito (SonarCloud); defined acceptance thresholds and escalation protocols
- Enforced security compliance using Snyk and StackHawk; documented APIs with Swagger/OpenAPI
- Managed **high-throughput production queues** with zero-downtime tolerance, prioritizing tasks under time pressure
- CI/CD deployments via Git, Azure DevOps, and Gitflow; monitored via Lens, Azure Logs, and KQL
- Maintained full **audit trail documentation** for all processes and operational procedures
- Developed Bash automation scripts to update client's data

Founder & Full Stack Developer

Pura Fruta — E-commerce Entrepreneurship *Nov 2020 – May 2021*
Quito, Ecuador

- Launched e-commerce delivery platform during COVID-19, achieving **$25K in sales**
- Built and maintained business website using MERN stack (MongoDB, Express, React, Node.js)
- Developed custom order management system in Python (pandas, NumPy)
- Configured and managed DigitalOcean production server

INTERNSHIPS & ACADEMIC ROLES

Teaching Assistant — Physics 2

Yachay Tech University *Apr 2020 – Jun 2020*
Ecuador

- Graded homework and laboratory assignments; provided tutoring and student support

Practicing Programmer

ATBS - Talov *Mar 2021 – Jun 2021*
Ibarra, Ecuador

- Generated 900 sign language videos for apps Speakliz and Vision, serving the deaf community
- Applied 10× multiplication pipeline to scale the video dataset

FabLab Technician (Volunteer)

Yachay Tech University — Universidad de Investigación *Oct 2016 – Jul 2018*
Ecuador

- Prototype construction and operation of 3D printers, laser cutters, and CNC machines
- Maintained equipment and developed safe operating procedures for shared lab tools

TECHNICAL SKILLS

Nanoscience & Lab	SPPS · Drug encapsulation · HPLC · Raman · AFM (contact mode, force spectroscopy, nanolithography) · TEM/HRTEM/SAED · ESI-MS · Fluorographene exfoliation · Optical microscopy
Instrumentation & Metrology	AFM (WSxM, force curve analysis) · Digital Micrograph · SAED pattern indexing · d-spacing measurement · Raman spectroscopy · HPLC/UHPLC-MS · Arduino sensor platforms
Process & Data Analysis	Python (pandas, NumPy, Matplotlib, scipy) · Statistical process control (SPC) · IV curve fitting · FFT/IFFT image processing · RMSD/cluster analysis · Molecular dynamics (AMBER/VMD) · DFT/VASP · Mathematica · Hertz/Sneddon fitting
Software Engineering	Java · Spring Boot · Spring WebFlux · JUnit · Mockito · Python · JavaScript · React · Node.js · Express · HTML/CSS · Bash
Infrastructure & Tools	Git · GitHub · Azure DevOps · Gitflow · PostgreSQL · MongoDB · Redis · Salesforce · Swagger · SonarCloud · Snyk · StackHawk · Docker · CI/CD · Lens · KQL · Agile/Scrum
Fabrication & Prototyping	3D printing · Laser cutting · CNC machines
Documentation & Markup	LaTeX · Swagger/OpenAPI
Languages	Spanish (native) · English (professional proficiency) · Dutch (A2, actively learning)

CERTIFICATIONS

Build Reactive MicroServices using Spring WebFlux/SpringBoot — Udemy	*Aug 2023*
Learn Java Unit Test: Mockito & Powermock Complete Tutorial — Udemy	*Oct 2023*
Reflections and Annotation in Java Uncomplicated — Udemy	*Sep 2023*
The Complete SSL and TLS Guide: HTTP to HTTPS — Udemy	*May 2023*
Introduction to Collections, Generics & Reflection in Java — Udemy	*Jan 2023*
The Complete Web Development Bootcamp — Udemy / Angela Yu	*Jan 2022*
Data Analysis with Python — freeCodeCamp	*Apr 2021*

CONGRESSES & EVENTS

Student Excellence Day — imec — Leuven, Belgium	*Oct 2025*
Advanced Functional Nanomaterials School — Yachay Tech University	*Feb 2020*
Acton University (leadership programme) — Michigan, USA	*Jun 2018*
Agro Tech Symposium — Quito, Ecuador	*Jul 2017*